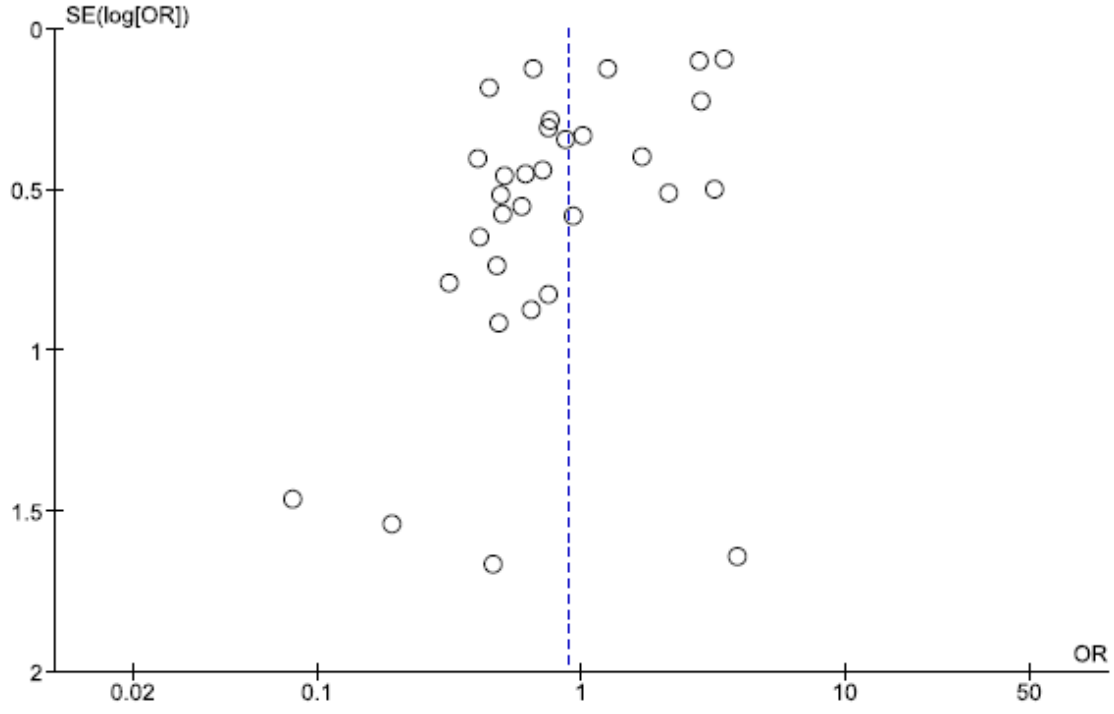
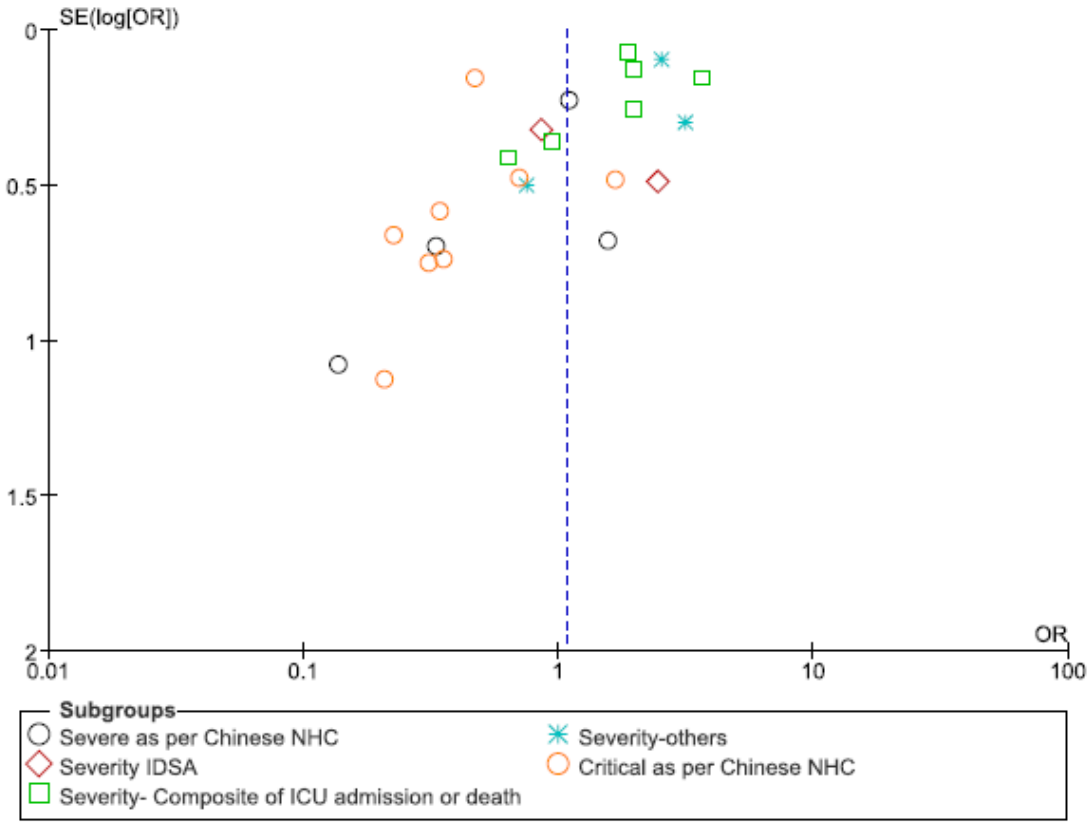
Supplementary Materials

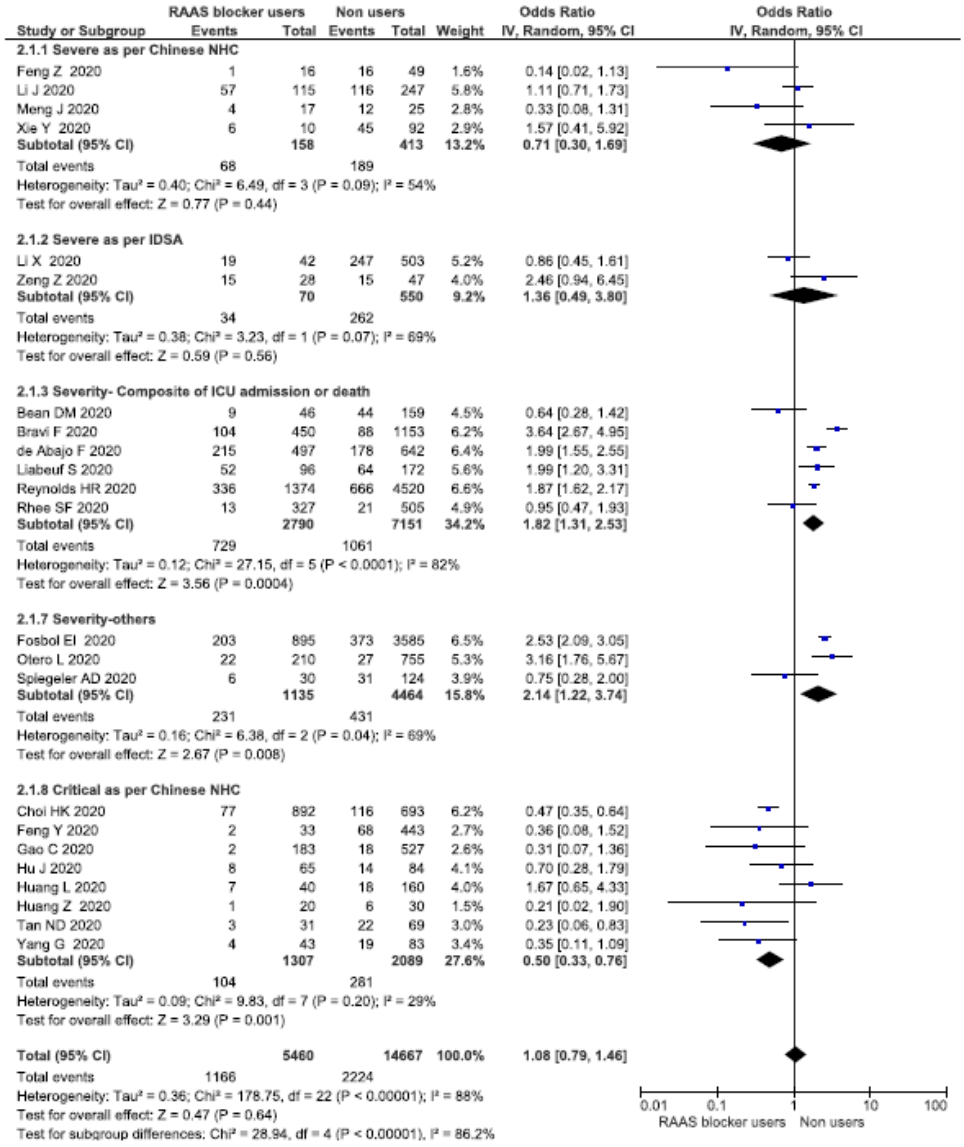
Supplementary Figure 1. Funnel plot for mortality analysis



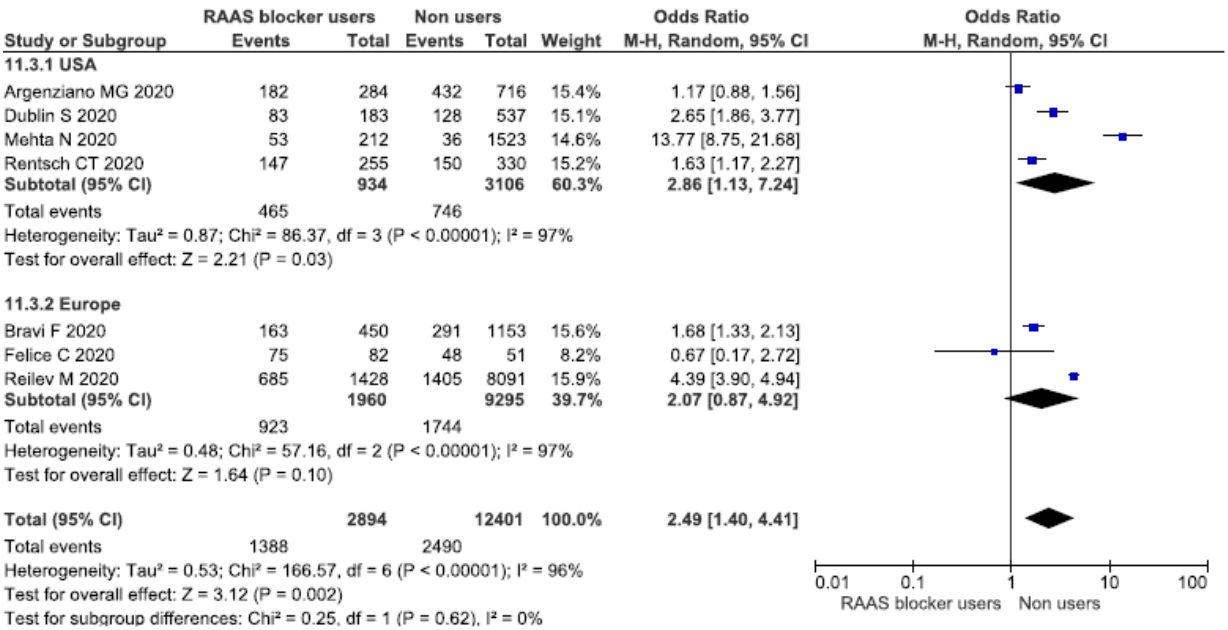
Supplementary Figure 2. Funnel plot for severity analysis



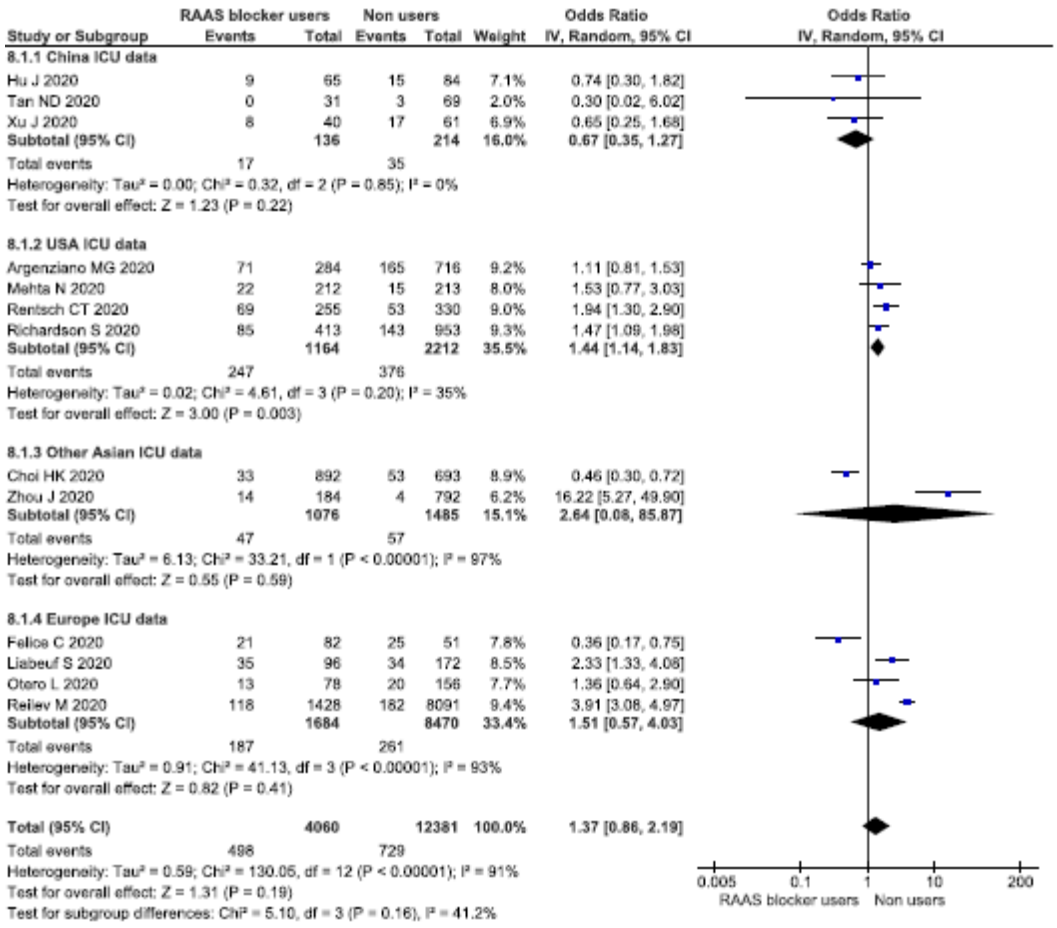
Supplementary Figure 3. Analysis of studies with respect to definition of severity



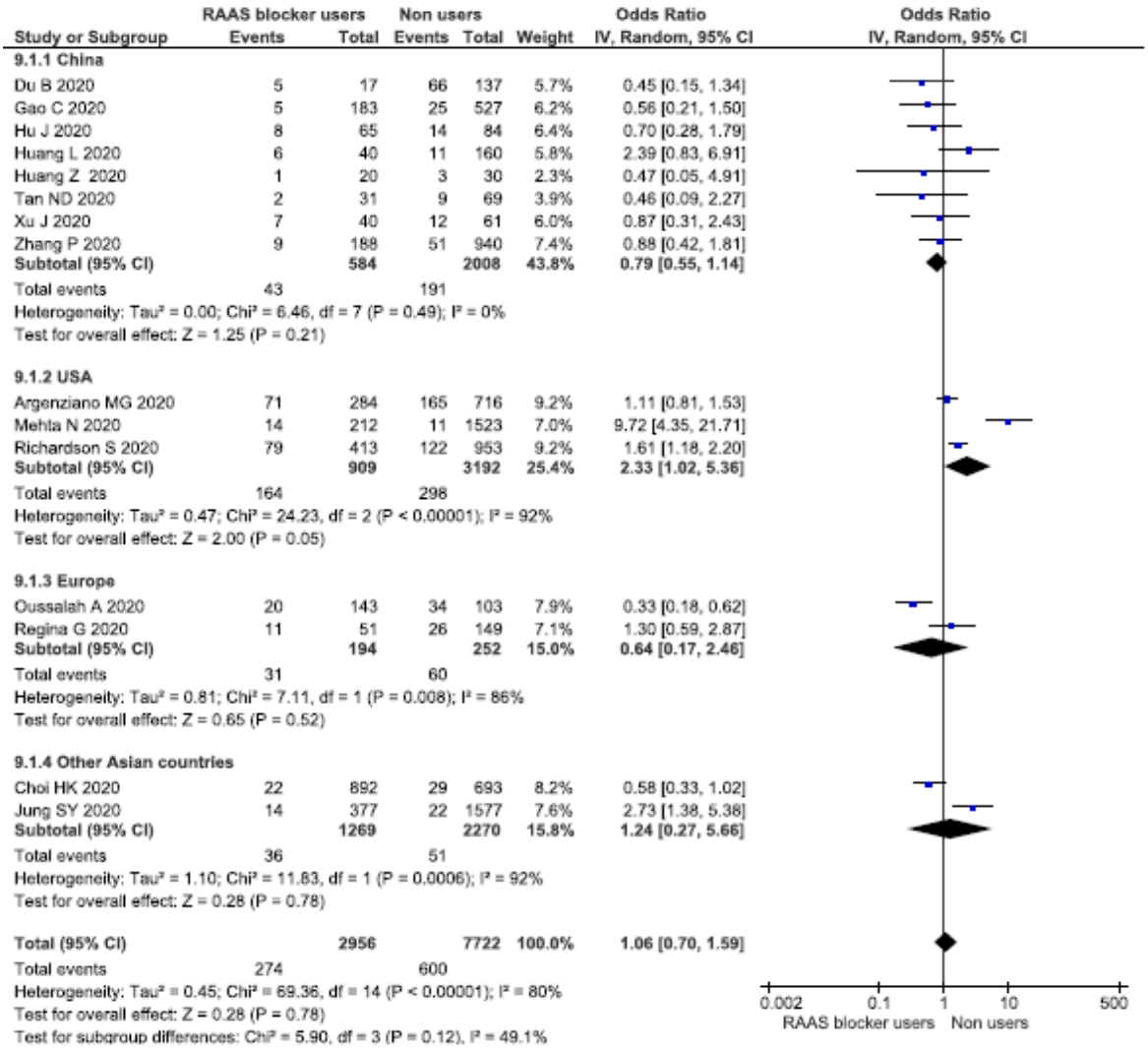
Supplementary Figure 4. Analysis of studies with respect to hospitalization



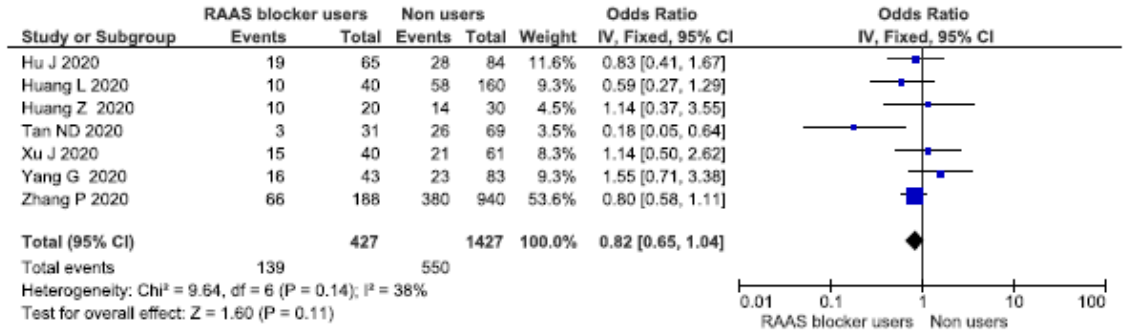
Supplementary Figure 5. Analysis of studies with respect to ICU admission



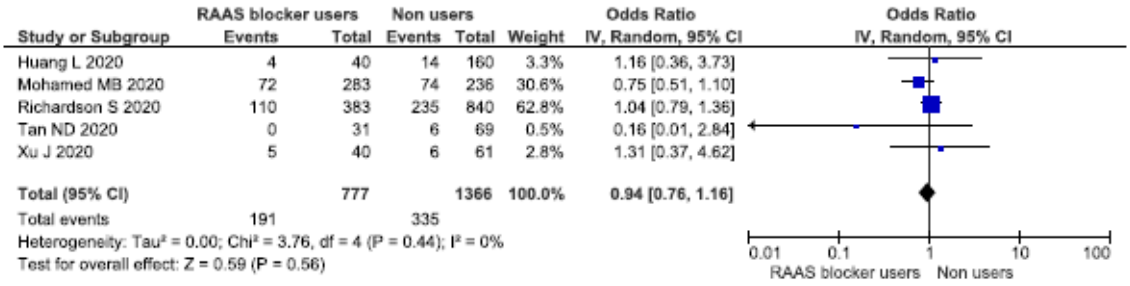
Supplementary Figure 6. Analysis of studies with respect to invasive ventilation



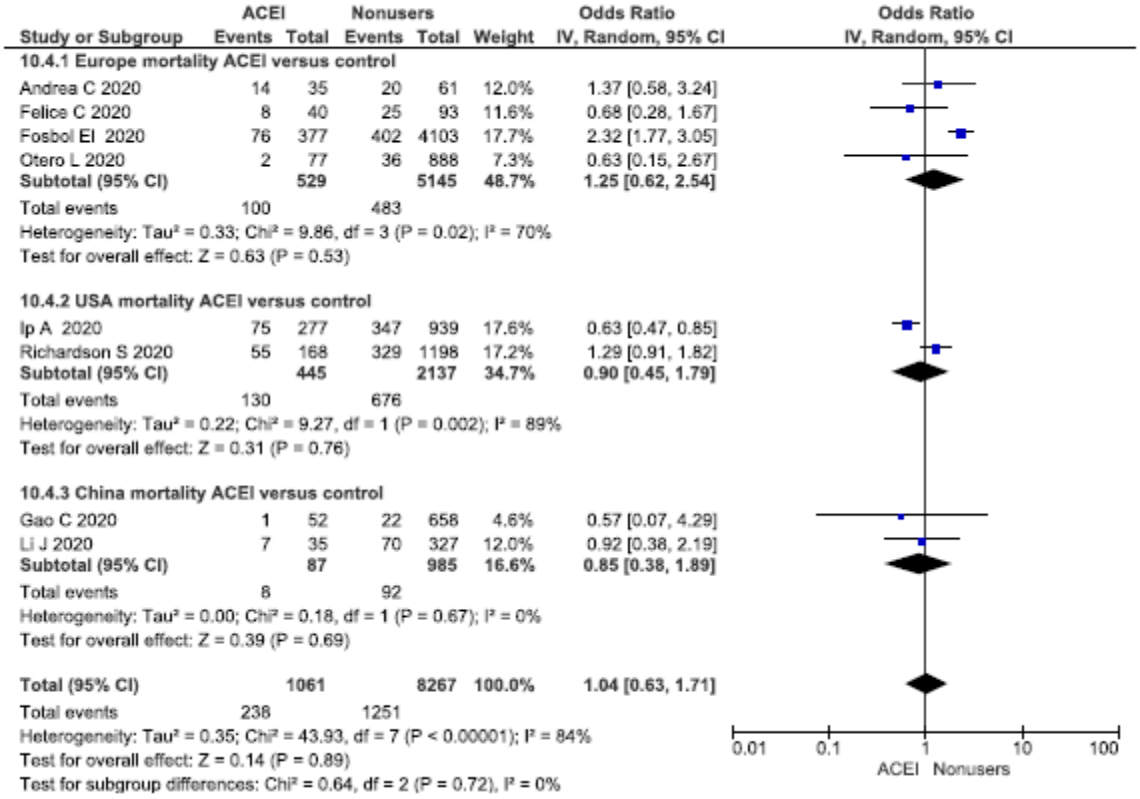
Supplementary Figure 7. Analysis of studies with respect to steroid use



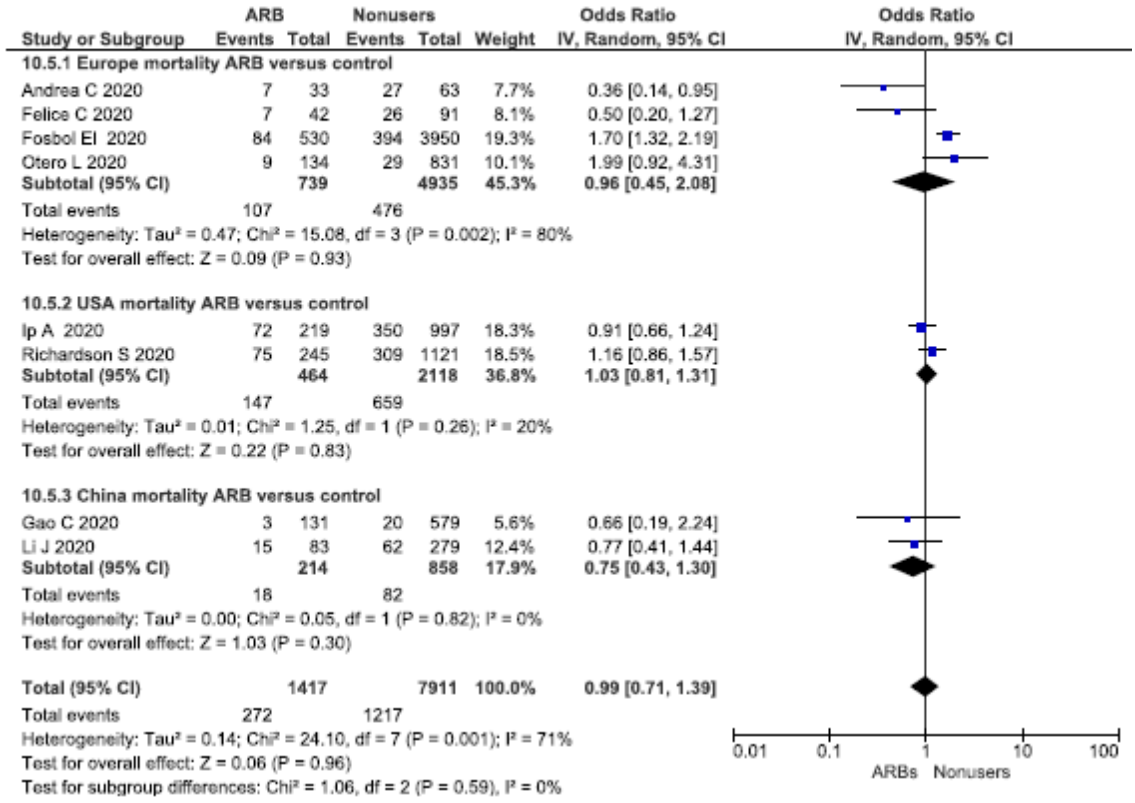
Supplementary Figure 8. Analysis of studies with respect to acute kidney injury



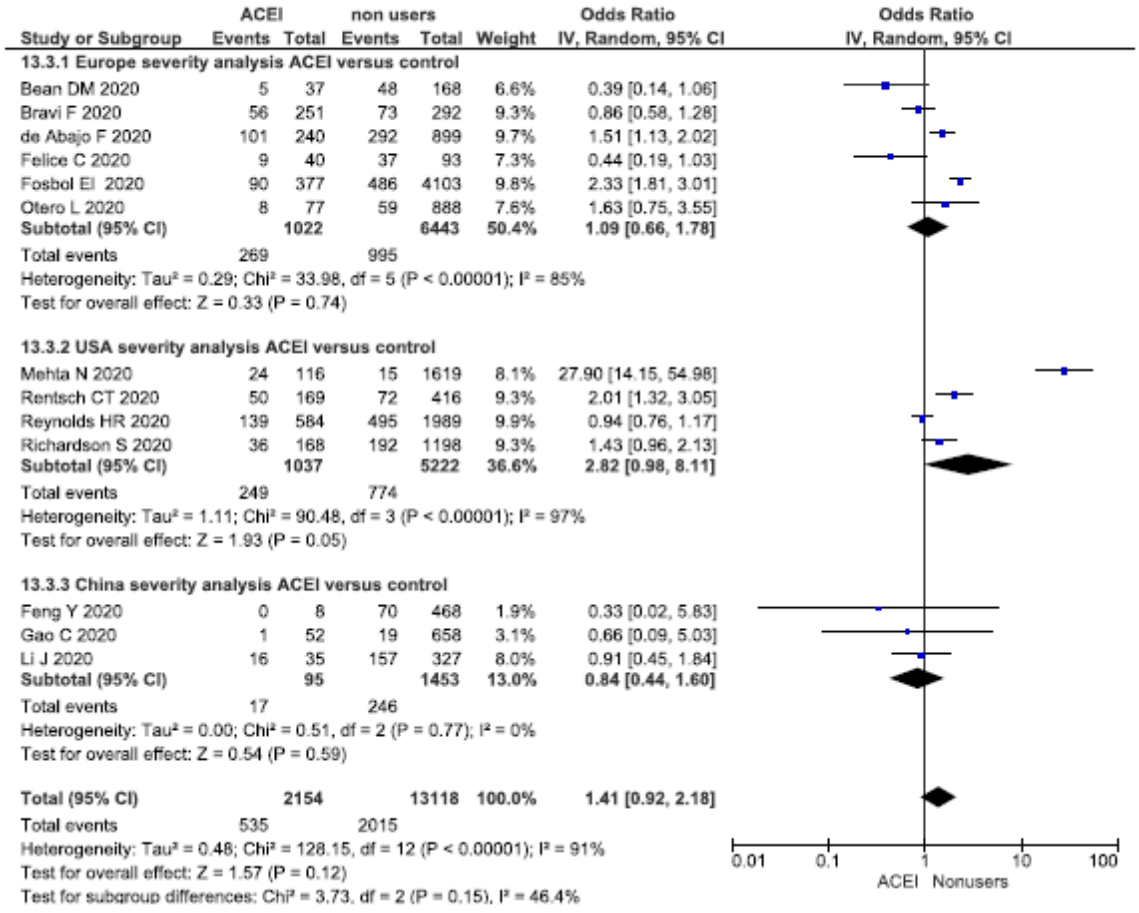
Supplementary Figure 9: Analysis of mortality outcomes with ACEIs



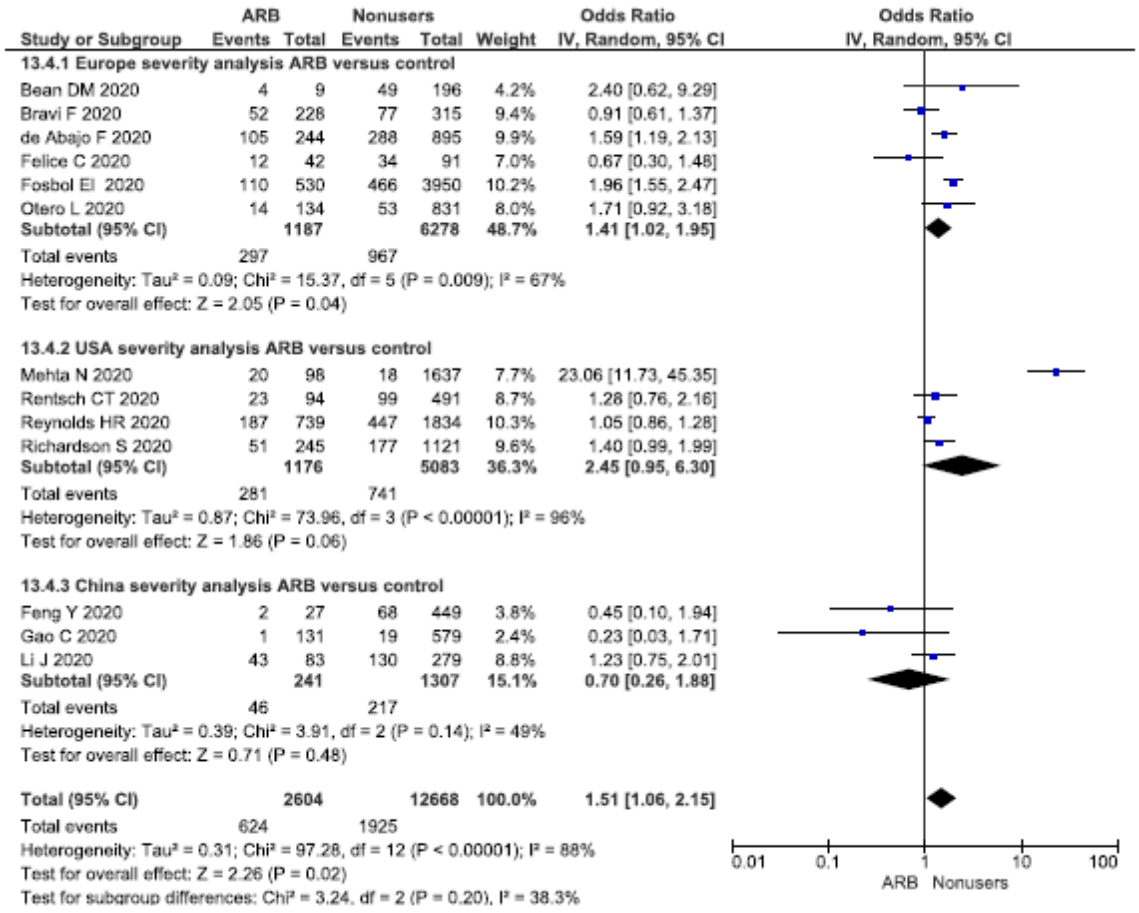
Supplementary Figure 10: Analysis of mortality outcomes with ARBs



Supplementary Figure 11: Analysis of severity outcomes with ACEI use



Supplementary Figure 12: Analysis of severity outcomes with ARBs



Supplementary Table 1:

Mortality analysis	**Number of studies (number of patients)**	**OR (CI)**	**OR (CI)Sensitivity analysis**
ACEI versus control	8(9328)	1.04 (0.63-1.71), I^2=84%	1.17 (0.6- 2.30), I^2=65%
ARB versus control	8(9328)	0.99 (0.71- 1.39), I^2=71%	0.92 (0.46- 1.81), I^2=76%
ACEI versus ARB	8(2478)	1.07 (0.88-1.32), I^2=22%	1.32 (0.98- 1.79), I^2=1%
Country wise mortality ACEI versus control	8(9328)	1.04 (0.63-1.71), I^2=84%	1.17 (0.6- 2.30), I^2=65%
Europe mortality	4(5674)	1.25 (0.62-2.54), I^2=70%	1.25 (0.61-2.55), I^2=70%
USA mortality	2(2582)	0.90 (0.45-1.79), I^2=89%	---
China mortality	2(1072)	0.85 (0.38-1.89), I^2=0%	0.57 (0.07- 4.29)
Country wise mortality ARB versus control	8(9328)	0.99 (0.71- 1.39), I^2=71%	0.92 (0.46- 1.81), I^2=76%
Europe mortality	4(5674)	0.96 (0.45-2.08) I^2=80%	0.96 (0.45-2.08) I^2=80%
USA mortality	2(2582)	1.03 (0.81- 1.31), I^2=20%	---
China mortality	2(1072)	0.75 (0.43- 1.30), I^2=0%	0.66 (0.19- 2.24)
Severity analysis	**Number of studies (number of patients)**	**OR (CI)**	**OR (CI)Sensitivity analysis**
ACEI vs control	N=13(15272)	**1.41** (0.92-2.18), I^2=91%	**1.45** (0.89-2.35), I^2=92%

Composite of ICU/death	4(4460)	0.98(0.68-1.40), I^2= 74%	0.98(0.68-1.40), I^2= 74%
ICU admission	4(3819)	2.45(0.65-9.18), I^2=96%	2.94(0.37-23.32), I^2=97%
'Critical' as per NHC	2(1186)	0.52(0.1-2.75), I^2=0%	0.66(0.09-5.03)
'Severe' as per NHC	1(362)	0.91(0.45-1.84)	--
Others*	2(5445)	2.25(1.77-2.87) I^2=0%	2.25(1.77-2.87) I^2=0%
ARBs vs control	N=13(15272)	**1.51** (1.06- 2.15), I^2=88%	**1.6** (1.07-2.40), I^2=89%
Composite of ICU/death	4(4460)	1.20 (0.89-1.62), I^2=63%	1.20 (0.89-1.62), I^2=63%
ICU admission	4(3819)	2.29 (0.64-8.13), I^2=95%	2.71 (0.34- 21.92), I^2=97%
'Critical' as per NHC	2(1186)	0.35 (0.11-1.16), I^2=0%	0.23 (0.03, 1.71)
'Severe' as per NHC	1(362)	1.23 (0.75-2.01)	--
Others*	2(5445)	1.93(1.55-2.39), I^2=0%	1.93 (1.55- 2.39) I^2=0%
ACEI versus ARB	N=13(4758)	1.00 (0.87- 1.14), I^2=0%	0.99(0.86-1.15), I^2=0%
Composite of ICU/death	4(2332)	0.92 (0.77- 1.11), I^2=18%	0.92 (0.77, 1.11), I^2=18%
ICU admission	4(972)	1.06 (0.78-1.45), I^2=0%	1.08 (0.73- 1.61), I^2=0%
'Critical' as per NHC	2(218)	1.34 (0.17-10.81), I^2=0%	2.55 (0.16, 41.53)
'Severe' as per NHC	1(118)	0.78 (0.35- 1.73), I^2=0%	--
Others*	2(1118)	1.17 (0.87-1.58), I^2=0%	1.17 (0.87- 1.58), I^2=0%

Country wise severity, ACEI versus control	N=13(15272)	**1.41** (0.92-2.18), I^2=91%	**1.45** (0.89-2.35), I^2=92%
Severity Europe	6(7465)	1.09 (0.66-1.78) , I^2=85%	1.09 (0.66- 1.78) , I^2=85%
Severity USA	4(6259)	**2.82** (0.98-8.11), I^2=97%	3.63 (0.76-17.33), I^2=98%
Severity China	3(1548)	0.84 (0.44- 1.60), I^2=0%	0.66 (0.09- 5.03)
Country wise severity ARB versus control	N=13(15272)	**1.51** (1.06- 2.15), I^2=88%	**1.6** (1.07-2.40), I^2=89%
Severity Europe	6(7465)	**1.41** (1.02-1.95), I^2=67%	**1.41** (1.02-1.95), I^2=67%
Severity USA	4(6259)	**2.45** (0.95- 6.33), I^2=96%	3.06 (0.64-14.65), I^2=97%
Severity China	3(1548)	0.69 (0.25- 1.90), I^2=51%	0.23 (0.03- 1.71)

*Study by Otero DL et al assessed severity as composite of heart failure/death, study by Fosbol EL et al assessed severity as SARS/ICU admission
ACEI: Angiotensin converting enzyme inhibitor, ARB: Angiotensin receptor blocker, CI: Confidence interval, NHC: National Health Commission of China, OR: Odds ratio

Supplementary Table 1: Mortality and severity outcomes, definition wise and geographic location wise in COVID-19 patients on ACEIs and ARBs.